PROJECT AND TEAM INFORMATION

Project Title

HyperVision: A Resource-Aware DBMS–OS Laboratory Using Xen and Virtual Machine Introspection

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Database performance does not depend only on the SQL query.It can also be affected by other things like CPU,memory,I/O,threads and the virtual machine where the database is running.It can be difficult to study database and operating system together because they are usually studied as separate topics.Our project HyperVision tries to connect these topics and understand how they affect the database perfomance.In this project,we will make a simple database system and run different tasks inside a Xen virtual machine.While the tasks are running,we will collect information fromthe database,operating system and virtual machine.We will also use Virtual Machine Introspection (VMI) to collect some information from the virtual machine without directly working inside it.The main aim of HyperVision is to find the reason why the database becomes slow.It will check if the problem is related to CPU,memory,I/O or how the computer resources are being used.After collecting the information,HyperVision will sugest some changes in the way resources are given to the system. Then the same task will be run again to see if the performance is improvd.

State of the Art

Nowadays,database performance is mainly checked from the database side.It tells us how much time a query takes, how it runs, how many rows it works on and some other information. There are also some tools which check the operating system. They show things like CPU usage, memory usage,runing processes and I/O work. Some tools also check how the virtual machine is working.The main problem is that all this informtion is checked separtely. It is not properly connected with each other,so it can be hard to find out why the database is becoming slow. For example,if CPU usage becomes high when a query is running,it does not always mean that the query needs a lot of CPU.HyperVision tries to solve this problems by using a simple DBMS with Xen and VMI.It collects information from the database, operating system and virtual machine and keeps it together using time and experiment IDs.This helps us to understand the real reason for the slow performance.

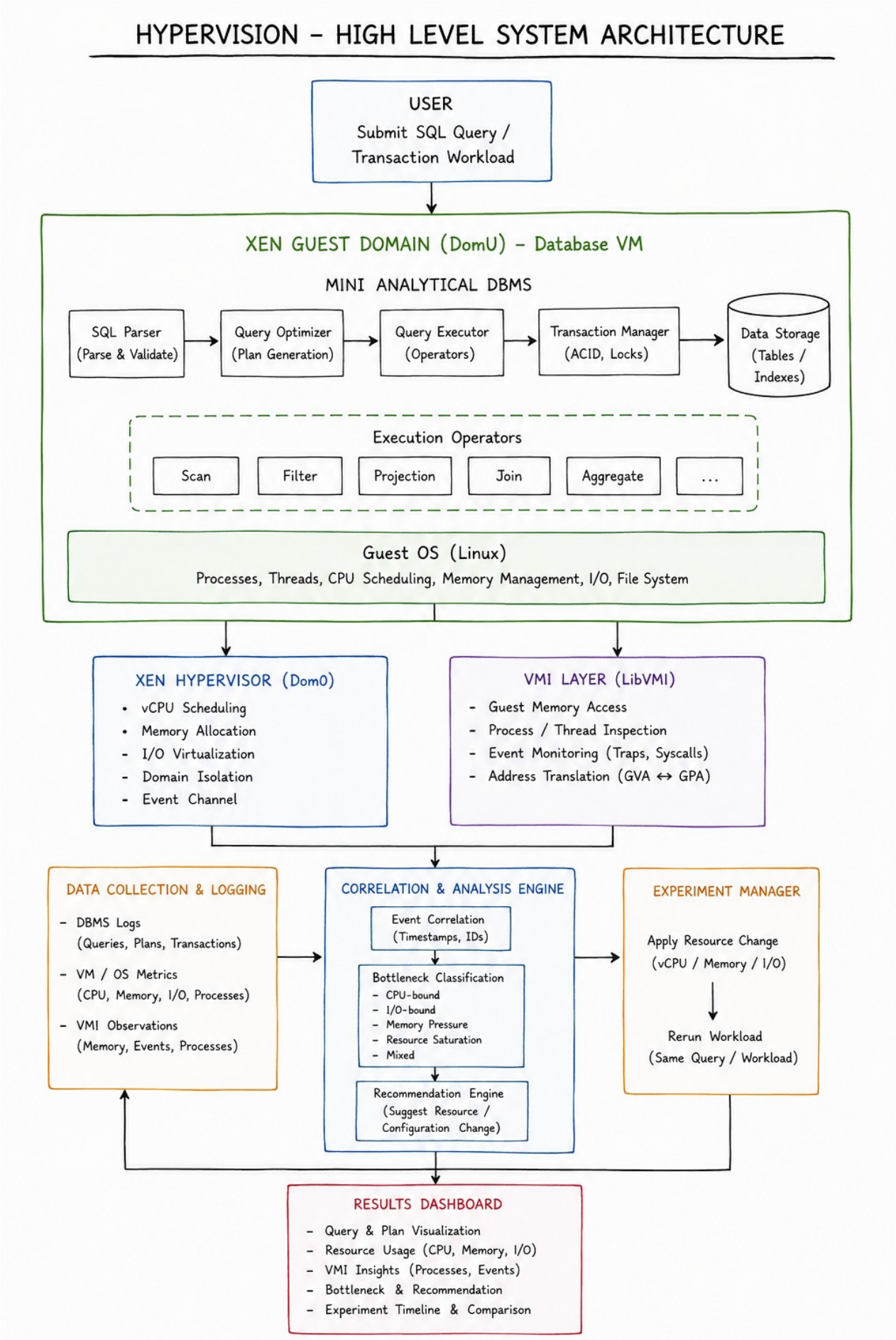
Project Goals and Milestones

The main aim of HyperVision is to make a simple system which help us understand how a database works and how it uses computer resources like CPU,memory and I/O.In this project,we will make a simple database system which can perform basic SQL operations like Scan,Filter,Projecton,Join and Group.The database work will be run inside a Xen virtual machine called DomU.We will use Xen and LibVMI to collect information from the virtual machine and its operating system. The information from database, virtual machine and VMI will be saved with same time and experiment ID. This will help us to compare the information from different parts of the system easily.HyperVision will find common problems related to CPU, memory and I/O. After finding the problem, it will suggest how the resources can be changed.Then same work will be run again to check if the performance is improved or not.At the end,we will make a dashboard where we can easily see the experiments and their results. The main work of this project includes making the DBMS,adding SQL parsing and query operations, setting up Xen and LibVMI,collecting information,comparing the data,finding problems and making the dasboard.

Project Approach

HyperVision will be developed step by step. First,we will do a small test using a simple database system made in Python,C or C++.The database will run SQL queries and do basic operations like scan, filter, project, hash join and group/aggregate.The database will run inside a Xen virtual machine called DomU. We will change and control resources like vCPUs and memory to see how different types of work affects the database.We will use LibVMI and a simple VMI layer to collect information about memory,processes,threads and other activities without directly working inside the guest operating system.All the information collected from the database, virtual machine and VMI will be saved with the experiment ID and time.We will check the type of problem,like CPU problem,I/O problem,memory problem,lack of resources or a mix of these problems.Bassed on the problem, HyperVision will suggest some changes and run the same work again.The new results will be compared with the old results to see if there is any improvement.Finally,the resultss will be shown on a simple web page made using HTML,CSS,JavaScript,Python and Flask.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The final project will be about making a working version of HyperVision.It will connect the database with the operating system and virtual machine so we can see how they work together.In this project,we will make a simple database system which can perform basic SQL operations like scan,filter,projection,hash join and group/aggregation.We will also set up the Xen environment and use DomU for running the database tasks.We will also set up LibVMI with a simple VMI system. It will help us collect some information from the guest operating system without directly working inside it.A tool will be made to record the database and virtual machine information with the time and experiment ID. Another tool will check if the problem is related to CPU, memory, I/O or lack of resources.After finding the problem,HyperVVision will suggest some changes in the resorces.The same task will be run again so we can compare the new result with the old one.Finally,we will make a simple web dashboard. It will show the task details,query plan,CPU and memory usage,VMI information,problems found,suggested changes and the results of the experiments.

Assumptions

We assume that this project will be developed on a Linux/x86-64 computer which supports hardware virtulization, so we can run Xen.The database systm will be small and will have only the features which are needed for our project and testing. We will collect only some required information from the guest system using VMI.We are not going to make the complete Xen hypervisor or complete LibVMI. We will use the existing source code of Xen and LibVMI and add only the parts which are needed for our project.During the testing, we will change only one compter resource at a time.This will help us to understands clearly how each resource affects the performance of the database.

References

1. Xen Project. Xen Project Documentation and User Handbook. Available at: https://xenproject.org/
2. Xen Project. Xen Project User Handbook – Building and Configuring Xen. Available at: https://handbook.xenproject.org/
3. Xen Project. Xen Hypervisor Documentation. Documentation for Xen architecture, domains, virtualization and resource management.
4. LibVMI. LibVMI Documentation. Documentation covering Virtual Machine Introspection, guest memory access and address translation. Available at: https://libvmi.com/